

Python Mastery: Core Foundations & Loops

Syntax, Variables, Collections, Logic, and Iteration

1. Python Loops

Python has two primitive loop commands: while loops and for loops.

The While Loop

With the while loop we can execute a set of statements as long as a condition is true.

```
i = 1
while i < 6:
    print(i)
    if i == 3:
        break # Exit loop even if condition is true
    i += 1
```

The For Loop

A for loop is used for iterating over a sequence (list, tuple, dictionary, set, or string).

```
fruits = ["apple", "banana", "cherry"]
for x in fruits:
    if x == "banana":
        continue # Skip the rest of this iteration
    print(x)
```

The Range() Function

To loop through a set of code a specified number of times, we can use the `range()` function.

```
for x in range(2, 30, 3):
    print(x) # Starts at 2, ends at 29, increments by 3
```

2. Nested Loops

A nested loop is a loop inside a loop. The "inner loop" will be executed one time for each iteration of the "outer loop".

```
adj = ["red", "big", "tasty"]
fruits = ["apple", "banana", "cherry"]

for x in adj:
    for y in fruits:
        print(x, y)
```